On the Lower Bound for the Minimax Rate of Causal Effect Estimation

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Abstract

We prove a low-smoothness minimax lower bound for pointwise conditional average treatment effect estimation in a strict submodel with exactly uniform covariates. The construction replaces punctured covariate support used in previous work by a smooth quadratic partition of unity.

1 Settings and statement of the theorem

We observe independent copies of

$$Z = (X, A, Y), \qquad X \in [0,1]^d, \quad A \in \{0,1\}, \quad Y \in \{0,1\}.$$

For a distribution P, write

$$\pi_P(x) = P(A = 1 \mid X = x), \qquad \mu_{a,P}(x) = E_P(Y \mid X = x, A = a),$$

and

$$\tau_P(x) = \mu_{1,P}(x) - \mu_{0,P}(x).$$

We consider the submodel class $\mathcal{F}$ in which

$$X \sim \mathrm{Unif}([0,1]^d), \qquad \pi \in \mathcal{H}^\alpha, \quad \mu_0 \in \mathcal{H}^\beta, \quad \tau \in \mathcal{H}^\gamma, \quad \pi \in [\epsilon, 1-\epsilon]$$

Throughout, $\mathcal{H}^r$ denotes the unit ball in Holder space.

We use the squared Hellinger convention

$$H^2(P,Q) = \int (\sqrt{p} - \sqrt{q})^2 \, d\nu = 2\{1 - \mathrm{Aff}(P,Q)\},$$

where $\mathrm{Aff}(P,Q) = \int \sqrt{pq}\, d\nu$.

Theorem 1 (Uniform-covariate lower bound)**.** *Let $x_0 \in (0,1)^d$ and set*

$$s = \frac{\alpha + \beta}{2}.$$

If

$$s < \frac{d/4}{1 + d/(2\gamma)},$$

then there is a constant $c > 0$, independent of n, such that

$$\inf_{\widehat{\tau}} \sup_{P} E_P \, |\widehat{\tau}(x_0) - \tau_P(x_0)| \geq c\, n^{-1/(1+d/(2\gamma)+d/(4s))}.$$

The supremum is over the binary-outcome uniform-covariate Holder model described above.

Remark 1. *The theorem removes the punctured density support of X in[1].*

We use the standard fuzzy-hypotheses lemma in proof ([3]). In the form needed here, if $P_\lambda, Q_\lambda \in \mathcal{P}$ are indexed by λ with prior ϖ, if

$$H^2\left(\int P_\lambda^n \, d\varpi(\lambda), \int Q_\lambda^n \, d\varpi(\lambda)\right) \le \eta < 2,$$

and if $\psi(P_\lambda) - \psi(Q_{\lambda'}) \ge \Delta$ for all λ, λ', then

$$\inf_{\widehat{\psi}} \sup_{P \in \mathcal{P}} E_P \left|\widehat{\psi} - \psi(P)\right| \ge c_\eta \Delta$$

for a constant $c_\eta > 0$.

2 Construction of the Fuzzy Hypothesis

2.1 Quadratic Partition of Unity

We first construct the deterministic functions used in the hypotheses.

Lemma 1 (Smooth quadratic partition of unity)**.** *There exists $\phi \in C_c^\infty(\mathbb{R}^d)$ such that*

$$\sum_{m \in \mathbb{Z}^d} \phi(u - m)^2 = 1, \qquad u \in \mathbb{R}^d.$$

Proof. Choose $\rho \in C_c^\infty(\mathbb{R})$, $\rho \not\equiv 0$, such that

$$\sum_{k \in \mathbb{Z}} \rho(u - k)^2 > 0 \qquad \text{for all } u \in \mathbb{R}.$$

For example, one may take a nonnegative smooth bump which is positive on $(-1, 1)$. Define

$$\phi_1(u) = \frac{\rho(u)}{\left\{\sum_{k \in \mathbb{Z}} \rho(u - k)^2\right\}^{1/2}}.$$

The denominator in this expression is a smooth, strictly positive, periodic function of u. Hence $\phi_1 \in C_c^\infty(\mathbb{R})$ and

$$\sum_{k \in \mathbb{Z}} \phi_1(u - k)^2 = 1.$$

Set $\phi(u) = \prod_{j=1}^d \phi_1(u_j)$. Tensorization gives the claimed identity. □

Choose $G \in C_c^\infty(\mathbb{R}^d)$ such that $0 \le G \le 1$, $G(0) = 1$, and G is supported in $[-\frac{1}{10}, \frac{1}{10}]^d$. Since x_0 is an interior point of $[0, 1]^d$, for all $h < 1$,

$$\operatorname{supp}(G_h) \subset [0, 1]^d,$$

Where $G_h(x) = G\left(\frac{x - x_0}{h}\right)$.

Let

$$g_h(x) = G_h(x)^2.$$

For $\ell \le h$, define

$$\psi_m(x) = G_h(x)\phi\left(\frac{x - x_0}{\ell} - m\right), \qquad U_\lambda(x) = \sum_m \lambda_m \psi_m(x),$$

where λ_m, $m \in \mathbb{Z}^d$, are independent Rademacher variables. In the sum defining U_λ, only active indices m for which ψ_m is not identically zero need be retained. Their number is

$$M \asymp (h/\ell)^d.$$

The partition identity gives

$$\sum_m \psi_m(x)^2 = G_h(x)^2 \sum_m \phi\left(\frac{x - x_0}{\ell} - m\right)^2 = g_h(x). \tag{1}$$

For later use define

$$J(x) = \{m : \psi_m(x) \neq 0\}.$$

Since ϕ is compactly supported, there are constants $N_0, R_0 < \infty$ such that

$$|J(x)| \le N_0 \tag{2}$$

for all x, and, if $J(x) \cap J(x') \neq \varnothing$, then

$$x, x' \in \operatorname{supp}(G_h), \qquad \left\|x - x'\right\|_\infty \le R_0 \ell. \tag{3}$$

Let

$$a = c_\pi \ell^\alpha, \qquad b = c_\mu \ell^\beta, \qquad \tau_h(x) = 4ab\, g_h(x),$$

where $c_\pi, c_\mu > 0$ will be chosen small enough to guarantee positivity.

The fuzzy hypotheses are as follows, it is easy to verify that they all lies in the submodel class $\mathcal{F}$.

Case I: $\alpha \ge \beta$**.** Define P_λ by

$$\pi_P(x) = \frac{1}{2}, \qquad \mu_{0,P}(x) = \frac{1}{2} + bU_\lambda(x) - \frac{\tau_h(x)}{2}, \qquad \tau_P(x) = \tau_h(x),$$

and define Q_λ by

$$\pi_Q(x) = \frac{1}{2} + aU_\lambda(x), \qquad \mu_{0,Q}(x) = \frac{1}{2} + bU_\lambda(x), \qquad \tau_Q(x) = 0.$$

Case II: $\beta \ge \alpha$**.** Define P_λ by

$$\pi_P(x) = \frac{1}{2} + aU_\lambda(x), \qquad \mu_{0,P}(x) = \frac{1}{2} - \frac{\tau_h(x)}{2}, \qquad \tau_P(x) = \tau_h(x),$$

and define Q_λ by

$$\pi_Q(x) = \frac{1}{2} + aU_\lambda(x), \qquad \mu_{0,Q}(x) = \frac{1}{2} + bU_\lambda(x), \qquad \tau_Q(x) = 0.$$

In both cases the marginal distribution of X is exactly uniform on $[0,1]^d$.

3 One-observation mixture equality

Let R be the reference law under which $X \sim \mathrm{Unif}([0,1]^d)$ and A, Y are independent fair coins. Write

$$r = 2A - 1, \qquad t = 2Y - 1.$$

Likelihood ratios below are with respect to R.

Proposition 1 (Exact one-observation equality)**.** *In both regimes $\alpha \geq \beta$ and $\beta \geq \alpha$,*

$$\int P_\lambda \, d\varpi(\lambda) = \int Q_\lambda \, d\varpi(\lambda)$$

as one-observation distributions.

Proof. First suppose $\alpha \geq \beta$. Under P_λ,

$$\mu_{A,P}(x) = \frac{1}{2} + bU_\lambda(x) + \frac{r\tau_h(x)}{2}.$$

Thus

$$\rho_\lambda^P(z) = 1 + 2bt\, U_\lambda(x) + rt\, \tau_h(x) = 1 + 2bt\, U_\lambda(x) + 4ab\, rt\, g_h(x).$$

Under Q_λ,

$$\rho_\lambda^Q(z) = 4\left(\frac{1}{2} + raU_\lambda(x)\right)\left(\frac{1}{2} + tbU_\lambda(x)\right),$$

so

$$\rho_\lambda^Q(z) = 1 + 2ar\, U_\lambda(x) + 2bt\, U_\lambda(x) + 4ab\, rt\, U_\lambda(x)^2.$$

Using $\mathbb{E}_\lambda U_\lambda(x) = 0$ and (1),

$$\mathbb{E}_\lambda \rho_\lambda^P(z) = 1 + 4ab\, rt\, g_h(x) = \mathbb{E}_\lambda \rho_\lambda^Q(z).$$

Now suppose $\beta \geq \alpha$. Under P_λ,

$$\mu_{A,P}(x) = \frac{1}{2} + \frac{r\tau_h(x)}{2},$$

and hence

$$\rho_\lambda^P(z) = 4\left(\frac{1}{2} + raU_\lambda(x)\right)\left(\frac{1}{2} + \frac{tr\tau_h(x)}{2}\right).$$

Since $\tau_h = 4abg_h$,

$$\rho_\lambda^P(z) = 1 + 2ar\, U_\lambda(x) + 4ab\, rt\, g_h(x) + 8a^2 b\, t\, U_\lambda(x) g_h(x).$$

Under Q_λ,

$$\rho_\lambda^Q(z) = 1 + 2ar\, U_\lambda(x) + 2bt\, U_\lambda(x) + 4ab\, rt\, U_\lambda(x)^2.$$

Taking expectation over the Rademacher signs gives

$$\mathbb{E}_\lambda \rho_\lambda^P(z) = 1 + 4ab\, rt\, g_h(x) = \mathbb{E}_\lambda \rho_\lambda^Q(z).$$

The equality of likelihood ratios after mixing proves the result. □

4 Upper Bound for the Hellinger Distance

Let

$$\mathbb{P}_n = \int P_\lambda^n \, d\varpi(\lambda), \qquad \mathbb{Q}_n = \int Q_\lambda^n \, d\varpi(\lambda).$$

Lemma 2 (Finite-overlap product-mixture bound)**.** *There are constants $c_0, C > 0$ such that, if $n\ell^d \le c_0$, then*

$$H^2(\mathbb{P}_n, \mathbb{Q}_n) \le Cn^2 h^d \ell^d (a^2 b^2 + a^4), \qquad \alpha \ge \beta,$$

and

$$H^2(\mathbb{P}_n, \mathbb{Q}_n) \le Cn^2 h^d \ell^d (a^2 b^2 + b^4), \qquad \beta \ge \alpha.$$

Consequently, in either case,

$$H^2(\mathbb{P}_n, \mathbb{Q}_n) \le Cn^2 h^d \ell^d \ell^{4s}.$$

The proof occupies the rest of this section.

4.1 Conditional factorization

Condition on $X_1, \dots, X_n$ and write $W_i = (A_i, Y_i)$. Build a graph $\mathcal{G}_X$ on $\{1, \dots, n\}$ by connecting i and j when

$$J(X_i) \cap J(X_j) \neq \varnothing.$$

Let $\mathcal{C}_X$ denote its connected components. For a component $C \in \mathcal{C}_X$, set

$$M_C = \bigcup_{i \in C} J(X_i).$$

If $C \neq C'$, then $M_C \cap M_{C'} = \varnothing$; otherwise some sign would belong to both unions, creating an edge between the two components. Thus the signs involved in distinct components are independent.

Let $\rho_i^P = \rho_\lambda^P(X_i, A_i, Y_i)$ and $\rho_i^Q = \rho_\lambda^Q(X_i, A_i, Y_i)$. Relative to fair coins on the W_i's, the conditional mixture densities are

$$\bar{p}_X(w_1, \dots, w_n) = \mathbb{E}_\lambda \prod_{i=1}^n \rho_i^P, \qquad \bar{q}_X(w_1, \dots, w_n) = \mathbb{E}_\lambda \prod_{i=1}^n \rho_i^Q.$$

Independence of the signs across different connected components gives

$$\bar{p}_X(w_1, \dots, w_n) = \prod_{C \in \mathcal{C}_X} \bar{p}_C(w_C), \qquad \bar{q}_X(w_1, \dots, w_n) = \prod_{C \in \mathcal{C}_X} \bar{q}_C(w_C),$$

where

$$\bar{p}_C(w_C) = \mathbb{E}_\lambda \prod_{i \in C} \rho_i^P, \qquad \bar{q}_C(w_C) = \mathbb{E}_\lambda \prod_{i \in C} \rho_i^Q.$$

If $C = \{i\}$ is a singleton, Proposition 1 yields

$$\bar{p}_C(w_i) = \bar{q}_C(w_i).$$

4.2 From conditional to unconditional Hellinger distance

The X^n-marginal under both $\mathbb{P}_n$ and $\mathbb{Q}_n$ is iid uniform. Therefore

$$H^2(\mathbb{P}_n, \mathbb{Q}_n) = \mathbb{E}_X H_X^2(\bar{P}_X, \bar{Q}_X), \tag{4}$$

where H_X denotes Hellinger distance between the conditional laws of $W_1, \ldots, W_n$ given $X_1, \ldots, X_n$.

For product measures, Hellinger affinity tensorizes:

$$\mathrm{Aff}\left(\bigotimes_C P_C, \bigotimes_C Q_C\right) = \prod_C \mathrm{Aff}(P_C, Q_C).$$

Since $H^2(P, Q) = 2\{1 - \mathrm{Aff}(P, Q)\}$,

$$H_X^2(\bar{P}_X, \bar{Q}_X) = 2\left[1 - \prod_{C \in \mathcal{C}_X} \left\{1 - \frac{1}{2} H^2(\bar{P}_C, \bar{Q}_C)\right\}\right].$$

Using $1 - \prod_j (1 - u_j) \le \sum_j u_j$ for $u_j \in [0, 1]$, and using the singleton equality,

$$H_X^2(\bar{P}_X, \bar{Q}_X) \le \sum_{C \in \mathcal{C}_X : |C| \ge 2} H^2(\bar{P}_C, \bar{Q}_C). \tag{5}$$

4.3 Walsh-polynomial component bound

We use the language of Walsh charater theory in our proof([2]). Fix a component C with $r = |C|$, and fix both X_C and $w_C = (r_i, t_i)_{i \in C}$, where $r_i = 2A_i - 1$ and $t_i = 2Y_i - 1$.

For a finite set of signs M_C, define the Walsh character

$$\chi_S(\lambda) = \prod_{m \in S} \lambda_m, \qquad S \subseteq M_C.$$

Every sign polynomial has an expansion

$$F(\lambda) = \sum_{S \subseteq M_C} \widehat{F}(S) \chi_S(\lambda).$$

Define the Walsh ℓ_1 norm

$$\|F\|_W = \sum_S |\widehat{F}(S)|.$$

Then

$$|\mathbb{E}_\lambda F| \le \|F\|_W, \qquad \|FG\|_W \le \|F\|_W \|G\|_W. \tag{6}$$

Lemma 3 (Elementary Walsh cancellation)**.** *Let M be a finite set of Rademacher signs and, for $S \subseteq M$, write*

$$\chi_S(\lambda) = \prod_{m \in S} \lambda_m.$$

Then, for any $S_1, \ldots, S_k \subseteq M$,

$$\mathbb{E}_\lambda \prod_{j=1}^k \chi_{S_j}(\lambda) \ne 0 \quad \Longleftrightarrow \quad S_1 \triangle \cdots \triangle S_k = \varnothing,$$

where $\triangle$ denotes symmetric difference. In that case the expectation equals 1.

Proof. The product $\prod_{j=1}^{k} \chi_{S_j}$ is the Walsh character

$$\chi_{S_1 \triangle \cdots \triangle S_k}.$$

The expectation of χ_S under independent Rademacher signs is 1 if $S = \varnothing$ and 0 otherwise. This proves the equivalence. □

For $i \in C$, write $J_i = J(X_i)$, $U_i = U_\lambda(X_i)$, and $g_i = g_h(X_i)$. Then

$$U_i = \sum_{m \in J_i} \psi_m(X_i) \chi_{\{m\}},$$

so finite overlap implies

$$\|U_i\|_W \le C. \tag{7}$$

Moreover,

$$V_i = U_i^2 - g_i = 2 \sum_{\substack{m<m' \\ m,m' \in J_i}} \psi_m(X_i)\psi_{m'}(X_i)\chi_{\{m,m'\}},$$

and hence

$$\|V_i\|_W \le C. \tag{8}$$

All monomials in U_i have degree one, while all monomials in V_i have degree two and no constant part. We shall use Lemma 3 below to identify which expanded products can have nonzero expectation.

Lemma 4 (Component bound, case $\alpha \ge \beta$)**.** *If $\alpha \ge \beta$, then for every component C of size r,*

$$H^2(\bar{P}_C, \bar{Q}_C) \le C^r (a^2 b^2 + a^4).$$

Proof. In this case write

$$\rho_i^P = 1 + B_i + T_i, \qquad B_i = 2bt_i U_i, \qquad T_i = 4abr_i t_i g_i,$$

and

$$\rho_i^Q = \rho_i^P + D_i, \qquad D_i = A_i + C_i,$$

where

$$A_i = 2ar_i U_i, \qquad C_i = 4abr_i t_i V_i.$$

The factors A_i and B_i are linear Walsh polynomials with Walsh norms $O(a)$ and $O(b)$, respectively. The factor C_i is a centered quadratic Walsh polynomial with Walsh norm $O(ab)$, and T_i is deterministic with absolute value $O(ab)$.

Expand

$$\bar{q}_C - \bar{p}_C = \sum_{\varnothing \ne S \subseteq C} \mathbb{E}_\lambda \left[\prod_{i \in S} D_i \prod_{j \in C \setminus S} \rho_j^P \right]. \tag{9}$$

Then expand every D_i as either A_i or C_i, and every ρ_j^P as 1, B_j, or T_j. This produces at most C^r terms. By (6), the absolute expectation of any expanded term is bounded by the product of the Walsh norms of its factors.

Consider an expanded term with nonzero expectation. It contains at least one difference factor. By Lemma 3, the symmetric difference of the Walsh supports appearing in the term must be empty.

If the term contains an A_i, then it contains a linear Walsh factor with coefficient of order a. The parity rule forces at least one additional nonconstant Walsh factor. If this additional factor

is linear, then its coefficient is of order a or b, giving a coefficient product bounded by $a^2 + ab$. If the additional factor is quadratic, then one linear factor and one centered quadratic factor still have zero expectation by Lemma 3; hence at least one further nonconstant factor is required. The resulting coefficient product is bounded by $a \cdot ab$, times an additional factor bounded by one, and is therefore bounded by $a^2 + ab$. Deterministic T-factors only multiply by powers of $ab \le 1$. Thus every nonzero term containing an A_i has coefficient product at most $C^r(a^2 + ab)$.

If the term contains no A_i, then it must contain at least one C_i, since a difference factor is present. The factor C_i is centered quadratic with coefficient of order ab. By Lemma 3, a single centered quadratic factor cannot survive alone, and it cannot survive together with only one linear factor. Hence a nonzero term either contains another centered quadratic factor, giving coefficient at most $(ab)^2 \le ab$, or contains at least two linear B-factors, giving coefficient $(ab)b^2 = ab^3 \le ab$. Additional factors contribute only further powers of a, b, or ab, all bounded by one after choosing c_π, c_μ sufficiently small.

Summing over the at most C^r expanded terms gives the uniform bound

$$|\bar{q}_C(w_C) - \bar{p}_C(w_C)| \le C^r(ab + a^2). \tag{10}$$

By positivity, each $\rho_i^P \ge \kappa > 0$ uniformly, and so

$$\bar{p}_C(w_C) = \mathbb{E}_\lambda \prod_{i \in C} \rho_i^P \ge \kappa^r.$$

Therefore, with expectation below taken under fair coins on W_C,

$$\chi^2(\bar{Q}_C, \bar{P}_C) = E_{\text{fair}} \frac{(\bar{q}_C - \bar{p}_C)^2}{\bar{p}_C} \le C^r(ab + a^2)^2 \le C^r(a^2b^2 + a^4).$$

Since $H^2 \le \chi^2$, the claim follows. □

Lemma 5 (Component bound, case $\beta \ge \alpha$)**.** *If $\beta \ge \alpha$, then for every component C of size r,*

$$H^2(\bar{P}_C, \bar{Q}_C) \le C^r(a^2b^2 + b^4).$$

Proof. In this case write

$$\rho_i^P = 1 + A_i + T_i + S_i,$$

where

$$A_i = 2ar_iU_i, \qquad T_i = 4abr_it_ig_i, \qquad S_i = 8a^2b\, t_iU_ig_i.$$

Furthermore,

$$\rho_i^Q = \rho_i^P + D_i, \qquad D_i = B_i + C_i + E_i,$$

with

$$B_i = 2bt_iU_i, \qquad C_i = 4abr_it_iV_i, \qquad E_i = -8a^2b\, t_iU_ig_i.$$

Thus A_i is linear of size a, B_i is linear of size b, S_i and E_i are linear of size a^2b, C_i is centered quadratic of size ab, and T_i is deterministic of size ab.

As in (9),

$$\bar{q}_C - \bar{p}_C = \sum_{\varnothing \ne S \subseteq C} \mathbb{E}_\lambda \left[\prod_{i \in S} D_i \prod_{j \in C \setminus S} \rho_j^P \right],$$

and we expand each D_i into B_i, C_i, E_i and each ρ_j^P into $1, A_j, T_j, S_j$.

Consider an expanded term with nonzero expectation. By Lemma 3, the symmetric difference of its Walsh supports must be empty. If the term contains a linear difference factor B_i or E_i, the parity rule forces at least one further nonconstant Walsh factor. If that further factor is linear, the product of the two relevant linear coefficients is one of

$$ba, \quad b^2, \quad b(a^2b), \quad (a^2b)a, \quad (a^2b)b, \quad (a^2b)^2,$$

all of which are bounded by $ab + b^2$ because $a, b \le 1$. If the further factor is quadratic, then one linear factor and one centered quadratic factor still have zero expectation by Lemma 3; hence at least one additional nonconstant factor is required, and the resulting coefficient product is again bounded by $ab + b^2$. Deterministic factors only contribute powers bounded by one.

If the term contains no linear difference factor, then it must contain at least one C_i, with coefficient of order ab. By Lemma 3, a single centered quadratic factor cannot survive alone or with only one linear factor. Therefore a nonzero term either contains another centered quadratic factor, giving coefficient at most $(ab)^2 \le ab$, or contains at least two linear factors from the P-side, each of size a or a^2b, giving coefficient at most $ab \cdot a^2 = a^3b \le ab$. These contributions are bounded by $ab + b^2$.

There are at most C^r expanded terms, and hence

$$|\bar{q}_C(w_C) - \bar{p}_C(w_C)| \le C^r(ab + b^2). \tag{11}$$

Positivity gives $\bar{p}_C(w_C) \ge \kappa^r$. Thus

$$\chi^2(\bar{Q}_C, \bar{P}_C) \le C^r(ab + b^2)^2 \le C^r(a^2b^2 + b^4).$$

Again $H^2 \le \chi^2$, proving the lemma. □

4.4 Counting nontrivial components

Let N_r be the number of connected components of $\mathcal{G}_X$ of size r.

Lemma 6 (Component counting)**.** *For every constant $C_0 > 0$, there is $C > 0$ such that*

$$\mathbb{E}_X \sum_{C \in \mathcal{C}_X : |C| = r} C_0^r \le C^r n^r h^d \ell^{d(r-1)}, \qquad r \ge 2.$$

Consequently, if $n\ell^d \le c_0$ for c_0 sufficiently small, then

$$\mathbb{E}_X \sum_{C \in \mathcal{C}_X : |C| \ge 2} C_0^{|C|} \le C n^2 h^d \ell^d.$$

Proof. It suffices to count connected vertex sets of size r, which may overcount components. Every connected set of $r \ge 2$ labelled vertices contains a spanning tree. Fix a labelled tree T on vertices $\{i_1, \dots, i_r\}$ and choose a root. If all tree edges are present in $\mathcal{G}_X$, then at least one edge is present. For any edge (i, j), the condition $J(X_i) \cap J(X_j) \ne \varnothing$ implies, by (3), that both X_i and X_j lie in $\mathrm{supp}(G_h)$. This set has volume $O(h^d)$. For each child vertex, the edge to its parent implies, again by (3), that the child lies in a cube of volume $O(\ell^d)$ around the parent. Since the X_i are iid uniform,

$$P_X\{T \text{ is realized}\} \le C h^d \ell^{d(r-1)}.$$

By Cayley's formula, there are r^{r-2} trees on r labelled vertices. Therefore

$$\mathbb{E}_X \sum_{C:\,|C|=r} C_0^r \le C_0^r \binom{n}{r} r^{r-2} C h^d \ell^{d(r-1)} \le C^r n^r h^d \ell^{d(r-1)}.$$

Summing over $r \ge 2$ gives

$$\mathbb{E}_X \sum_{C:\,|C|\ge 2} C_0^{|C|} \le C n^2 h^d \ell^d \sum_{r\ge 2} (Cn\ell^d)^{r-2}.$$

If $n\ell^d \le c_0$ and c_0 is sufficiently small, the geometric series is bounded. □

4.5 Proof of Lemma 2

Combine (4) and (5). In the case $\alpha \ge \beta$, Lemma 4 gives

$$H^2(\mathbb{P}_n, \mathbb{Q}_n) \le \mathbb{E}_X \sum_{C:\,|C|\ge 2} C^{|C|}(a^2b^2 + a^4).$$

Lemma 6 then yields

$$H^2(\mathbb{P}_n, \mathbb{Q}_n) \le C n^2 h^d \ell^d (a^2b^2 + a^4).$$

The case $\beta \ge \alpha$ is identical using Lemma 5:

$$H^2(\mathbb{P}_n, \mathbb{Q}_n) \le C n^2 h^d \ell^d (a^2b^2 + b^4).$$

Finally,

$$a^2b^2 \asymp \ell^{2\alpha+2\beta} = \ell^{4s}.$$

If $\alpha \ge \beta$ and $\ell \le 1$, then

$$a^4 \lesssim \ell^{4\alpha} \le \ell^{2\alpha+2\beta} \lesssim a^2b^2,$$

where constants absorb c_π, c_μ. If $\beta \ge \alpha$, then similarly

$$b^4 \lesssim \ell^{4\beta} \le \ell^{2\alpha+2\beta} \lesssim a^2b^2.$$

Thus in both regimes,

$$H^2(\mathbb{P}_n, \mathbb{Q}_n) \le C n^2 h^d \ell^d \ell^{4s}.$$

This completes the proof of the finite-overlap product-mixture lemma.

5 Final Minimax Lower Bound

Proof of Theorem 1. Choose ℓ and h so that

$$\ell^{2s} \asymp h^\gamma. \tag{12}$$

Equivalently,

$$\ell = h^{\gamma/(2s)}$$

up to a fixed multiplicative constant. Since $2s < \gamma$, this also ensures $\ell \le h$ for all sufficiently small h.

By Lemma 2,

$$H^2(\mathbb{P}_n, \mathbb{Q}_n) \le C n^2 h^d \ell^d h^{2\gamma}.$$

Choose h by

$$n^2 h^d \ell^d h^{2\gamma} \asymp 1. \tag{13}$$

Since $\ell^d = h^{\gamma d/(2s)}$, (13) becomes

$$n^2 h^{d+\gamma d/(2s)+2\gamma} \asymp 1.$$

Taking the γth power scale gives

$$h^\gamma \asymp n^{-1/(1+d/(2\gamma)+d/(4s))}. \tag{14}$$

It remains to verify the sparsity condition in Lemma 2. From (13),

$$n \asymp h^{-\gamma-d/2-\gamma d/(4s)}.$$

Therefore

$$n\ell^d \asymp h^{-\gamma-d/2-\gamma d/(4s)} h^{\gamma d/(2s)} = h^{\gamma d/(4s)-d/2-\gamma}.$$

This tends to zero if and only if

$$\frac{\gamma d}{4s} - \frac{d}{2} - \gamma > 0,$$

which is equivalent to

$$s < \frac{d\gamma}{4\gamma + 2d} = \frac{d/4}{1 + d/(2\gamma)}.$$

Thus the low-smoothness condition is exactly the condition under which the finite-overlap clusters are sparse.

The parameter separation is uniform over all λ, λ':

$$\tau_{P_\lambda}(x_0) - \tau_{Q_{\lambda'}}(x_0) = \tau_h(x_0) = 4abG(0)^2 \asymp \ell^{2s} \asymp h^\gamma.$$

Since $G(0) = 1$, (14) gives

$$\tau_h(x_0) \asymp n^{-1/(1+d/(2\gamma)+d/(4s))}.$$

The fixed multiplicative constant in the tuning (13) can be chosen so that the product-mixture Hellinger distance is bounded by a constant $\eta < 2$. The fuzzy-hypotheses lemma then gives

$$\inf_{\widehat{\tau}} \sup_{P} E_P \left|\widehat{\tau}(x_0) - \tau_P(x_0)\right| \ge c\, n^{-1/(1+d/(2\gamma)+d/(4s))}$$

for all sufficiently large n. The remaining finite set of smaller sample sizes can be absorbed by decreasing c, using any fixed two-point submodel with positive separation at x_0. This proves the theorem. □

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